

Symbiosis Institute of Technology, Nagpur

INDUSTRY INTERNSHIP REPORT

Submitted in partial fulfillment of requirement for the award of degree of

Bachelor of Technology
in
Computer Science and Engineering

by

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May 2026



**Symbiosis International (Deemed University),
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

DECLARATION

I, hereby declare that the Industry Internship report submitted herein has been carried out by me in **Solar Industries India Limited, Nagpur** towards partial fulfillment of requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering. The work is original and has not been submitted earlier as a whole or in part for the award of any degree / diploma at this or any other Institution / University.

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CERTIFICATE

This is to certify that the Industry Internship Report, titled **“Data Control and Validation Centre & AI Demand Sensing Platform,”** has been carried out under our supervision at **Solar Industries India Limited** by **Arushi Shivhare** in partial fulfillment of the requirements for the Bachelor of Technology in Computer Science and Engineering. The work submitted is comprehensive, complete and fit for evaluation.

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I am deeply grateful to my industry mentor, **Mr. Soham Das** (AIML Team Lead), for his constant guidance, insightful feedback, and encouragement throughout the internship. His practical expertise in AI/ML systems, data engineering, and production-level deployment played a crucial role in shaping my learning and the outcomes of this project. I also extend my appreciation to my colleagues and team members at **Solar Industries India Limited** for their cooperation, assistance, and for making my internship a pleasant and productive experience.

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Lastly, I would like to acknowledge my family and friends for their unwavering support and encouragement during this period.

Thank you all for your invaluable contributions to my learning and professional growth.



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ABSTRACT

This report documents the Industry Internship carried out at Solar Industries India Limited (SIIL), Nagpur, under the mentorship of Mr. Soham Das (AIML Team Lead) from January 2026 to April 2026. The internship spanned 14 weeks and involved an intensive progression from foundational Artificial Intelligence and Machine Learning (AI/ML) concepts to the development of a large-scale, production-grade data intelligence platform. The major project developed during this internship is titled the “Data Control and Validation Centre & AI Demand Sensing Platform.” In large-scale industrial environments such as Solar Industries, decision-making is often hindered by “dirty data” and inconsistent forecasting. To address this, a Medallion Data Architecture (Bronze → Silver → Gold layers) was implemented that automatically cleans, validates, and audits millions of sales records spanning 2022 to 2026. The platform features a Self-Healing data pipeline that ensures 100% Data Uniqueness and Identity Validation before any analysis is performed. For demand sensing, a high-performance AI Ensemble consisting of five models — including XGBoost, LightGBM, Random Forest, Temporal Fusion Transformer (TFT), and N-BEATS — was built. These models operate in parallel to provide 7, 15, and 30-day sales projections with built-in confidence scoring. On the frontend, the system uses React.js to deliver an interactive dashboard with “Time-Travel” anchoring, allowing stakeholders to view exactly how predictions branch from historical actuals. By utilizing multi-threaded backend processing via Django and ThreadPoolExecutor, data retrieval latency was reduced from 10 seconds to under 1 second — a 90% improvement. The internship also involved structured weekly learning covering ML algorithms (SVM, Random Forest, XGBoost, LightGBM), Deep Learning, NLP, Large Language Models (LLMs), LangChain, LangGraph, LlamaIndex, AI Agents, RAG pipelines, Crew AI, and time series forecasting — all of which directly supported the project.

Keywords: *Artificial Intelligence, Machine Learning, Data Pipeline, Demand Forecasting, Ensemble Models, Data Validation, Time Series Analysis*

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1. INTRODUCTION TO COMPANY

1.1 About the Company

Solar Industries India Limited (SIIL) is one of India's most prominent and diversified manufacturing conglomerates, with its headquarters located in Nagpur, Maharashtra. Founded in 1995, the company has grown from a regional explosives manufacturer into a global enterprise with a strong presence across multiple high-impact sectors including mining, defence, infrastructure, and aerospace. Solar Industries is listed on both the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) of India, reflecting its stature as a publicly trusted and financially transparent organization.



Figure 1.1: Logo of Solar Industries

The company is recognized as one of the largest manufacturers of industrial explosives in India and has consistently ranked among the top players globally in its domain. With operations spanning more than 50 countries, Solar Industries serves a diverse clientele ranging from large mining corporations and government defence establishments to infrastructure companies and international exporters.

1.2 Historical Background

Solar Industries India Limited was established in 1995 with a primary focus on the manufacturing of industrial explosives and blasting accessories. Over the course of three decades, the company has undergone a significant transformation, diversifying its product portfolio and expanding its operational footprint both domestically and internationally.

In the early 2000s, Solar Industries began investing heavily in research and development to enhance the quality and safety of its explosive products. This commitment to innovation led to strategic expansions into the defence and aerospace sectors, where the company now manufactures ammunition, rocket propellants, and defence-grade pyrotechnic systems. The company's listing on major stock exchanges marked a pivotal milestone, enabling it to attract large-scale investment and accelerate its expansion strategy.

By the 2020s, Solar Industries had firmly established itself not only as a manufacturing giant but also as a technology-forward enterprise, integrating smart manufacturing systems, ERP platforms, and AI-powered analytics into its operations. Today, the company's Information Technology division plays a critical supporting role in ensuring operational efficiency, data-driven decision-making, and global supply chain management.

1.3 Location

The corporate headquarters of Solar Industries India Limited is situated in Nagpur, Maharashtra, which serves as the central hub for its administrative, research, and technology operations. The company operates multiple manufacturing units spread across India, with facilities in Nagpur, Singrauli (Madhya Pradesh), Pune, and Vadodara, among others. Internationally, the company has established manufacturing and distribution units in Africa, Southeast Asia, the Middle East, and Australia, enabling it to serve global clients efficiently.

The company's Nagpur facility houses the central AIML and Data Science team, which was the primary site for the internship. This facility is equipped with modern computing infrastructure, high-performance servers, cloud connectivity through Amazon Web Services (AWS), and collaborative development environments that support agile software development and data engineering workflows.

1.4 Operational Structure

Solar Industries India Limited operates through a well-structured organizational hierarchy that supports both its core manufacturing business and its growing technology division. The company is led by its Board of Directors and senior management team, with distinct divisions managing operations, sales, finance, and technology.

The **Information Technology and AIML Division** is one of the most strategically important units within Solar Industries. This division is responsible for building and maintaining intelligent systems that support supply chain analytics, sales forecasting, ERP integration, and operational dashboards. The AIML team, under the leadership of the AIML Team Lead, develops and deploys machine learning models, data pipelines, and AI-driven tools that are used across the company's global operations.

Table 1.1: Organizational Structure of the AIML Division

Role	Responsibility
AIML Team Lead	Overall system design and AI strategy
Data Engineers	Pipeline architecture, ERP integration
ML Engineers	Model development, training, deployment
Frontend Developers	Dashboard design with React.js
DevOps Engineers	AWS infrastructure, CI/CD pipelines

1.5 Vision and Mission

Vision: To be a globally recognized leader in energetic materials and industrial solutions, driven by innovation, safety, and technology.

Mission: To deliver world-class explosives, ammunition, and technology-enabled solutions that empower industries and nations, while maintaining the highest standards of safety, sustainability, and customer satisfaction.

The company's vision for its AIML division specifically focuses on harnessing the power of artificial intelligence and data analytics to improve operational efficiency and reduce waste.

1.6 Products or Software Solutions

Solar Industries offers a broad range of products and services that span both its core manufacturing domain and its growing technology capabilities.

Core Manufacturing Products:

- **Industrial Explosives:** Bulk explosives, packaged explosives, emulsion explosives, and ANFO-based products used in open-pit mining, quarrying, and tunnelling.
- **Detonators and Initiating Systems:** Electric detonators, non-electric detonators, and electronic blasting systems for precision blasting operations.
- **Defence Products:** Ammunition, mortar shells, rocket propellants, bomb components, and pyrotechnic systems supplied to Indian and international defence forces.
- **Aerospace Propellants:** Solid propellants and composite formulations used in missiles, satellite launch vehicles, and rocket systems.

Technology and IT Solutions:

- **Automation Systems:** Smart factories using PLC/SCADA systems for real-time process control.
- **Data Analytics and AIML Platforms:** Machine learning pipelines, demand forecasting systems, and data validation dashboards.
- **ERP Systems:** SAP-based Enterprise Resource Planning for supply chain management and logistics.
- **Cybersecurity:** Systems for protecting sensitive defence and operational data.
- **R&D Software:** Simulation and testing tools used in the development of new explosive formulations and defence systems.

2. PROJECT WORK / CASE STUDY

2.1 Introduction

The major project undertaken during this internship is titled **“Data Control and Validation Centre & AI Demand Sensing Platform.”** This project was developed in response to a real operational challenge faced by Solar Industries India Limited: the company generates millions of sales records across its global operations every month, and ensuring the integrity, accuracy, and actionability of this data is a significant technical challenge.

In large-scale industrial environments, decisions related to production planning, inventory management, and supply chain optimization are heavily dependent on the quality of sales data. When data contains duplicates, missing entries, or inconsistencies — collectively referred to as “dirty data” — it leads to flawed business intelligence, inaccurate demand forecasts, and expensive operational errors. This project was designed to solve this problem at scale.

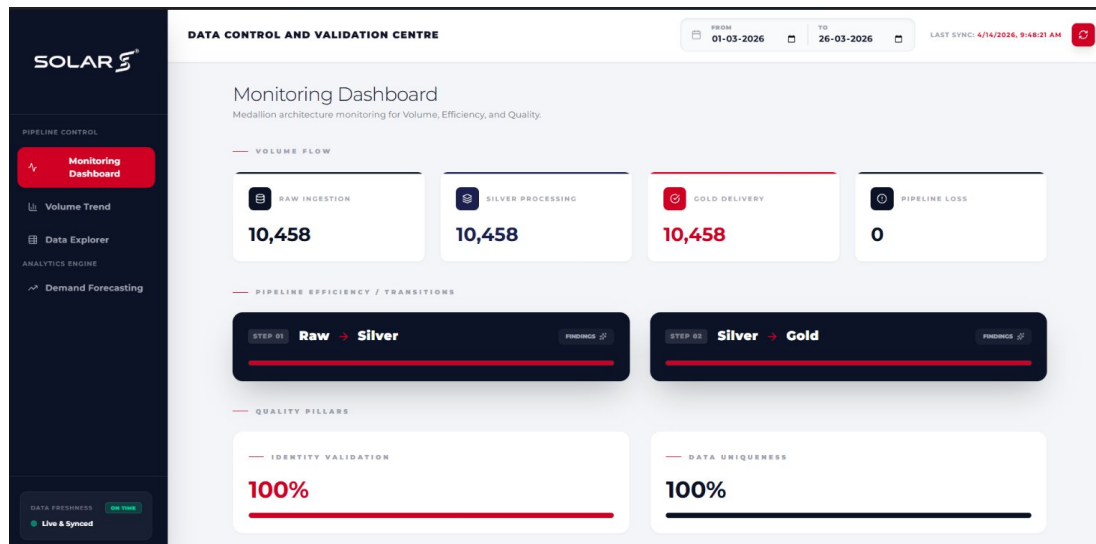


Figure 2.1: Data Control and Validation Centre – Monitoring Dashboard (Live System)

As shown in Figure 2.1, the Monitoring Dashboard provides a real-time view of the Medallion pipeline, showing raw ingestion counts, silver processing counts, gold delivery counts, pipeline loss, quality pillars (Identity Validation and Data Uniqueness), and SLA/data freshness status.

2.2 Problem Identification

The core problems addressed by this project are:

1. **Data Integrity Issues:** Millions of sales records pulled from the company’s ERP system (via AWS RDS) contained duplicate entries, missing identifiers, and formatting inconsistencies. These errors propagated through the analytics pipeline, leading to unreliable reports and misleading dashboards.
2. **Slow Dashboard Performance:** The existing data retrieval process required sequential database queries, resulting in dashboard loading times of up to 10 seconds. This latency made the tool impractical for live operational use.
3. **Inconsistent Demand Forecasting:** Business teams relied on manual spreadsheet-based forecasting for planning purposes. This approach was slow, error-prone, and unable to incorporate complex patterns in historical sales data such as seasonality, trend shifts, or customer-specific volatility.
4. **Lack of Data Transparency:** There was no system to audit how data was transformed from its raw form to its final analytical state, making it difficult for stakeholders to trust the outputs of any analysis.

2.3 Objective

The specific objectives of the project were:

1. To build a system that checks and cleans millions of company sales records, removing duplicates and correcting inconsistencies automatically.
2. To implement an AI-powered demand forecasting engine that predicts sales for specific items for 7, 15, and 30-day time horizons.
3. To prevent business decisions based on “bad” or broken data by instituting automated validation gates.
4. To create a live, self-healing dashboard that updates itself automatically and alerts users when data synchronization is delayed.
5. To cover sales data from 2022 to 2026 and support multi-product, multi-customer forecasting scenarios.
6. To achieve real-time performance through multi-threading so that the dashboard loads within 1 second.

2.4 Work Carried Out

2.4.1 Weekly Learning Progression

The internship followed a structured weekly learning plan that built competencies progressively, starting from foundational AI/ML concepts and moving toward advanced production-level system design. The following table summarizes the key activities and outcomes across the internship weeks.

Table 2.1: Weekly Internship Activities and Learning Outcomes

Week	Topic / Activity	Key Outcomes
1–6	ML Fundamentals, NLP, Deep Learning	Supervised/Unsupervised ML, neural networks, transformers
7	LLMs, AI Agents, Prompt Engineering	Understood LLM architecture, tokenization, chatbots, prompt chaining
8–9	RAG Pipelines, LangChain, LangGraph	Built retrieval-augmented generation systems and agentic workflows
10	LLM Fine-Tuning, Crew AI	Learned SFT, QLoRA, multi-agent AI architectures
11	LlamaIndex, Time Series Forecasting	Data indexing for LLMs, ARIMA, Prophet, XGBoost forecasting
12	SVM, Model Evaluation	Classification algorithms, accuracy, precision, recall, F1, RMSE
13	Claude AI, API Integration, MCP	AI API workflows, Model Context Protocol architecture
14	Data Pipelines, EDA, Forecasting Systems	End-to-end industrial data workflows and production system design

2.4.2 Large Language Models and AI Agents

During Week 7, a comprehensive study of Large Language Models (LLMs) and their applications was undertaken. Large Language Models are advanced AI systems trained on massive datasets to understand and generate human language. The core architecture behind modern LLMs is the **Transformer model**, which enables the system to analyze context within text and produce coherent responses.

Key concepts studied included tokenization (converting text to mathematical tokens), the training process (predicting the next word iteratively over large corpora), and the resulting language understanding capabilities — text generation, question answering, summarization, translation, and code generation.

AI Agents were studied as a natural extension of LLMs. Unlike simple chatbots, AI Agents combine LLMs with a reasoning engine, tool integration (APIs, databases, search engines), and memory systems. This combination enables agents to perform complex, multi-step tasks autonomously. Two primary development approaches were studied: framework-based development using LangChain and LangGraph, and direct API development for maximum flexibility.

Prompt Engineering was also studied as a critical skill for directing LLM behavior. Techniques included instruction-based prompting (providing explicit directives), prompt chaining (sequentially linking prompts for complex workflows), and prompt routing (directing different task types to specialized models).

2.4.3 LLM Fine-Tuning and Crew AI

Week 10 focused on advanced techniques for improving LLM performance. **Fine-tuning** involves adapting a pre-trained model to a specific domain using targeted datasets, rather than training from scratch. Two key methods were studied:

- **Supervised Fine-Tuning (SFT):** Training the model on labeled input-output pairs for controlled, accurate responses.
- **QLoRA (Quantized Low-Rank Adaptation):** A parameter-efficient technique that reduces memory and computational cost by updating only a small subset of model parameters.



Figure 2.2: LLM Fine-Tuning Pipeline

Crew AI was explored as a multi-agent paradigm where specialized agents (Research Agent, Writer Agent, Validator Agent) collaborate under a Manager Agent to complete complex workflows in parallel. This architecture enables research automation, multi-step reasoning, and enterprise-level AI workflows at scale.

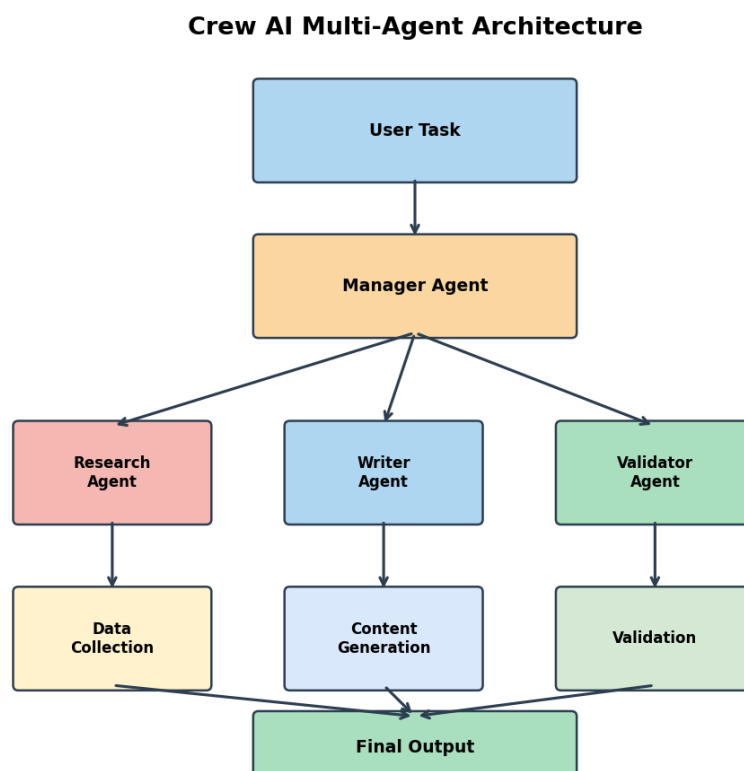


Figure 2.3: Crew AI Multi-Agent Architecture

2.4.4 LlamaIndex and Time Series Forecasting

LlamaIndex was studied as a framework that bridges external data sources with LLMs. Its workflow begins with raw data ingestion (from documents, databases, or APIs), followed by index creation and storage. A query engine then matches user queries to the most relevant data segments, which are passed to the LLM for context-aware response generation.

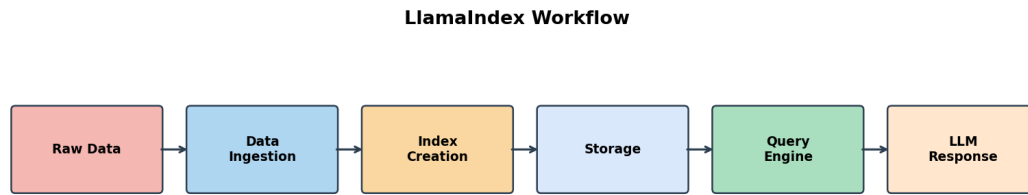


Figure 2.4: LlamaIndex Data Framework Workflow

Time Series Forecasting was studied in depth, given its direct relevance to the demand sensing project. Time series datasets consist of four components: *Trend* (long-term direction), *Seasonality* (repeating fixed-interval patterns), *Cyclic* (non-fixed fluctuations), and *Noise* (random variation). Traditional models like ARIMA and SARIMA handle linear patterns, while modern approaches like XGBoost and Prophet address non-linear relationships with higher accuracy.

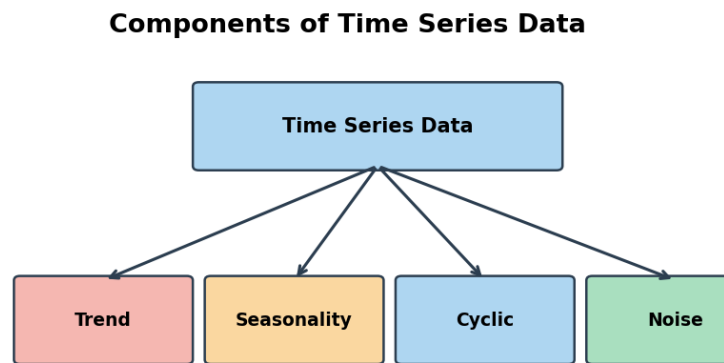


Figure 2.5: Components of Time Series Data

2.4.5 Support Vector Machine and Model Evaluation

Support Vector Machine (SVM) was studied as a powerful supervised learning algorithm for classification and regression. SVM finds the optimal *hyperplane* that maximizes the margin between classes. For non-linearly separable data, the **kernel trick** transforms data into a higher-dimensional space where linear separation becomes possible. The Radial Basis Function (RBF) kernel is the most commonly used variant for non-linear problems.

Model Evaluation metrics studied included:

- **Classification Metrics:** Accuracy, Precision, Recall, and F1-Score.
- **Regression Metrics:** RMSE (Root Mean Squared Error) and R^2 Score.
- **Cross-Validation:** Ensuring the model generalizes across different data subsets.

SVM Classification with Hyperplane

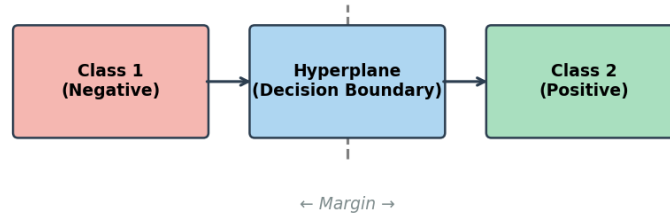


Figure 2.6: SVM Classification with Hyperplane Boundary

2.4.6 Claude AI, API Integration, and Model Context Protocol

Week 13 involved practical exposure to **Claude AI**, an advanced conversational AI system developed by Anthropic, used at Solar Industries for content generation, code debugging, and workflow automation. Key learning included understanding how Claude maintains contextual awareness, adapts output format based on instructions, and integrates with industry workflows.

API-Based AI Development was studied to understand how AI models are integrated into production applications. The workflow involves an application sending a request to an AI model through an API endpoint; the model processes the input, performs reasoning, and returns a response.

API-Based AI Integration Workflow

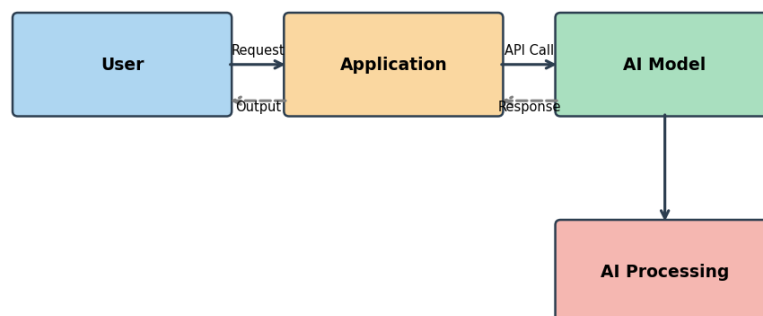


Figure 2.7: API-Based AI Integration Workflow

Model Context Protocol (MCP) was explored as a framework defining how AI models interact with external tools, manage memory, and maintain context. Components include Context Management (storing conversation history), Tool Integration (accessing external APIs or databases), and Query Handling (generating structured outputs).

Model Context Protocol (MCP) Architecture

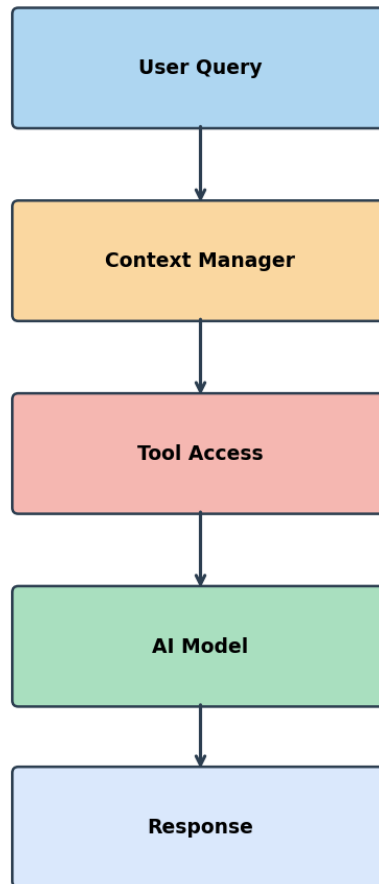


Figure 2.8: Model Context Protocol (MCP) Architecture

2.4.7 Data Pipelines, EDA, and Industry-Level Implementation

Week 14 synthesized all prior learning into a comprehensive understanding of industrial data workflows. The three-layer data pipeline (Raw → Silver → Gold) was studied in the context of the actual production system built at Solar Industries.

Data Validation was studied as a critical step involving checking for missing values, duplicate records, inconsistent formats, and outliers. Automated validation systems using error detection, debugging, and correction pipelines are used in production environments.

Exploratory Data Analysis (EDA) was performed on datasets to understand their structure, patterns, and relationships before applying machine learning models.

Industry-Level AI Deployment architecture was studied: the frontend handles user interaction, an API layer acts as a bridge, the backend manages logic and data processing, the ML model generates predictions, and all outputs are stored in a database.

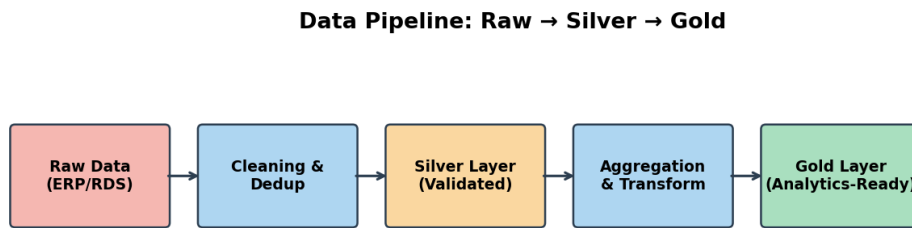


Figure 2.9: Three-Layer Data Pipeline: Raw → Silver → Gold

2.5 Solution Provided

2.5.1 Medallion Data Architecture

The foundation of the platform is a three-layer **Medallion Architecture** for managing and transforming raw ERP data into business-ready analytics:

- **Bronze Layer (Raw):** Unprocessed data pulled “as-is” from the company’s ERP system via AWS RDS. Contains all records regardless of quality.
- **Silver Layer (Validated):** Data is cleaned by removing duplicates, correcting missing names, fixing erroneous dates, and enforcing formatting standards. Identity Validation and Data Uniqueness scores are computed at this stage.
- **Gold Layer (Final):** Data is grouped by customer and product, aggregated, and made ready for AI model consumption and business reporting.

The transition findings between pipeline steps are visible to users through a popup interface, as shown in Figure 2.14.

2.5.2 AI Ensemble for Demand Sensing

The demand forecasting engine is built as an **ensemble of five AI models** running in parallel. The system automatically selects the best-performing model for each specific product-customer combination based on error scores.

2.5.3 Module Description

The platform is organized into four key functional modules:

The Live Database Explorer (Module 3) is shown in Figure 2.16, providing direct access to the Gold Layer and Silver Layer records with full date, customer, and product detail.

2.5.4 Technical Stack

2.5.5 Methodology

The system was built following a structured four-phase methodology:

Data Validation Flow

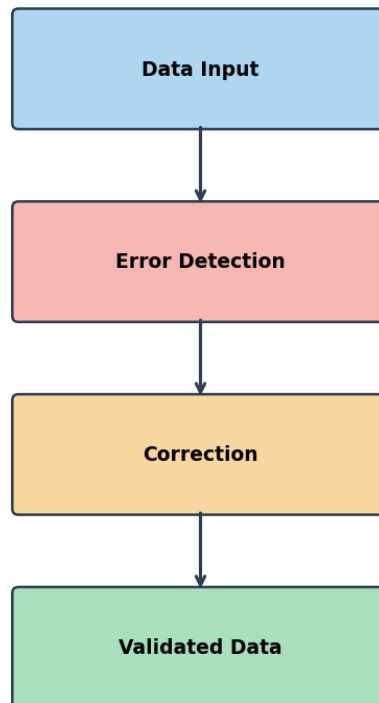


Figure 2.10: Data Validation Flow

Phase 1 – Data Organization (Medallion Architecture): Raw ERP sales data is ingested into the Bronze Layer. It is then cleaned — removing duplicates, fixing missing fields, and enforcing data type consistency — to produce the Silver Layer. Finally, the data is grouped by customer and product to produce the business-ready Gold Layer.

Phase 2 – Intelligence Building (AI Ensemble): Five AI models are trained on the Gold Layer data. The system runs all five models simultaneously (in parallel) and evaluates each using WAPE and R^2 metrics. The model with the lowest error score for each product-customer pair is automatically selected as the forecasting engine. A “Time-Travel” logic anchors the 30-day forecast to the latest available date in the database, ensuring always-current predictions.

Phase 3 – Performance Optimization: The Django backend uses Python to execute five database queries in parallel (multi-threading). This reduced dashboard loading latency from 10 seconds to under 1 second. A dual-database routing strategy is implemented to keep raw data queries separate from AI model queries, preventing them from competing for the same resources.

Phase 4 – Frontend Dashboard: A React.js interface was built displaying a solid line for historical actuals and branching colored lines for AI predictions across 7, 15, and 30-

Exploratory Data Analysis (EDA) Process

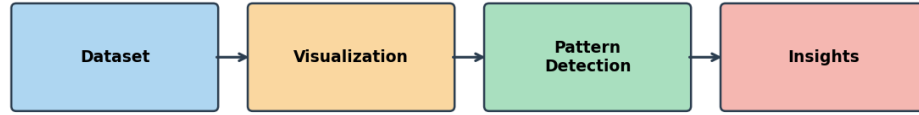


Figure 2.11: Exploratory Data Analysis (EDA) Process

Full Industry-Level AI System Architecture



Figure 2.12: Full Industry-Level AI System Architecture

day horizons. An SLA Health Check “traffic light” system alerts users to data freshness: Green (sync healthy), Yellow (sync delayed), Red (sync failed).

2.6 Results

2.6.1 System Performance

Figure 2.17 summarizes the six headline achievements of the platform.

The most significant technical achievement was the reduction in dashboard data retrieval latency from **10 seconds to under 1 second**, accomplished through multi-threaded backend parallelization using Python’s `ThreadPoolExecutor`. This 90% reduction in loading time makes the dashboard practical for real-time operational use.

2.6.2 Volume Trend Analysis

Figure 2.18 shows the System Volume Trend page, which displays the Ingestion vs. Delivery gap over the last 14 records. Key statistics include an Average Daily Volume of 8,355 records, a Peak Stress Load of 13,799 records (recorded on 03-01), and a Period Retention of 100.0%.

2.6.3 AI Demand Sensing Results

Figure 2.19 shows the AI Demand Sensing module’s model evaluation scorecard. Based on historical volatility analysis, **LightGBM** was automatically identified as the optimal forecasting engine for the selected customer segment, achieving a **59% Confidence Score**.

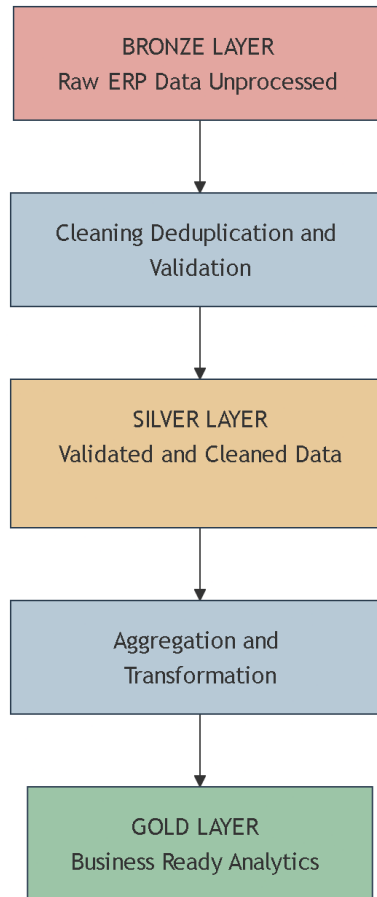


Figure 2.13: Medallion Data Architecture (Bronze → Silver → Gold)

Random Forest scored 48% and XGBoost scored 33%, confirming LightGBM's superiority for this dataset.

2.6.4 Demand Forecast Visualization

The demand forecast chart (Figure 2.20) shows two panels: the top panel displays 30 days of historical actuals (solid black line), and the bottom panel shows the next 30 days of AI predictions from all five ensemble models (LightGBM, NBEATS, Random Forest, TFT, XGBoost). Dotted lines indicate deep learning networks.

The forecasting pipeline followed to generate these results is shown in Figure 2.22.

2.7 Conclusion

The **Data Control and Validation Centre & AI Demand Sensing Platform** represents a complete, production-grade solution to the data quality and demand forecasting challenges faced by Solar Industries India Limited. Through the implementation of a Medallion Data Architecture, a five-model AI Ensemble, multi-threaded backend processing, and an interactive React.js dashboard, the platform delivers measurable and impactful results.

The platform ensures that all business decisions at Solar Industries are based on **100% clean, validated, and auditable data**. The AI ensemble provides accurate, horizon-

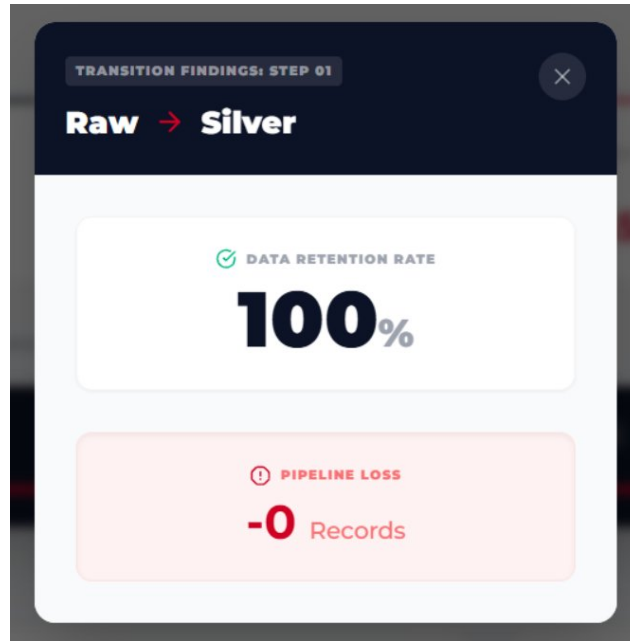


Figure 2.14: Transition Findings: Raw → Silver (100% Data Retention, 0 Pipeline Loss)

Table 2.2: AI Models in the Demand Sensing Ensemble

Model	Type	Strength
Random Forest	Classic Ensemble	Robust, handles non-linear patterns
LightGBM	Gradient Boosting	Fast, efficient, high accuracy
XGBoost	Gradient Boosting	High-performance, regularized
TFT (Temporal Fusion Transformer)	Neural (Deep Learning)	Long-range temporal patterns
N-BEATS	Neural	Interpretable time-series decomposition

specific demand forecasts that empower supply chain managers to plan production and inventory with confidence. The self-healing pipeline and SLA monitoring system ensure continuous reliability without manual intervention.

The internship also provided deep practical exposure to the broader AI/ML ecosystem — LLMs, RAG pipelines, AI Agents, Crew AI, fine-tuning, and time series forecasting — all of which contributed to the design and implementation of the final system. The experience of working on a real-world, production-level system at one of India’s most respected industrial companies has been transformative and has significantly strengthened.

AI Demand Sensing — Ensemble Architecture

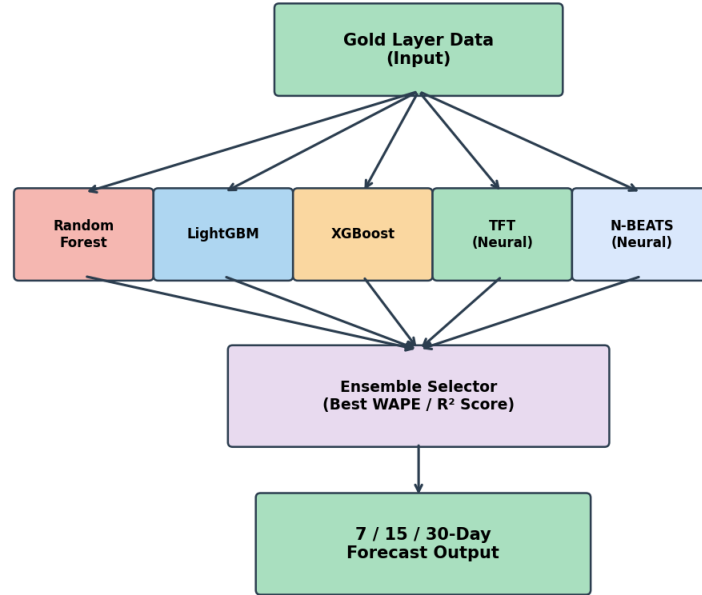


Figure 2.15: AI Demand Sensing Ensemble Architecture

Table 2.3: System Module Description

Module	Description
Module 1: Pipeline Control	Monitors real-time row counts across Bronze, Silver, and Gold layers. Tracks data retention rates and triggers SLA alerts for synchronization delays.
Module 2: Quality Engine	Calculates Identity Validation and Data Uniqueness scores. Automatically flags duplicate or incomplete records to ensure data integrity.
Module 3: Data Explorer	Provides an infinite-scroll table to audit specific records in each data layer. Enables CSV exports for manual engineering audits.
Module 4: AI Sensing	Visualizes 7/15/30-day demand forecasts using the ensemble of five ML models. Features “Time-Travel” anchoring to sync predictions with the latest actuals.

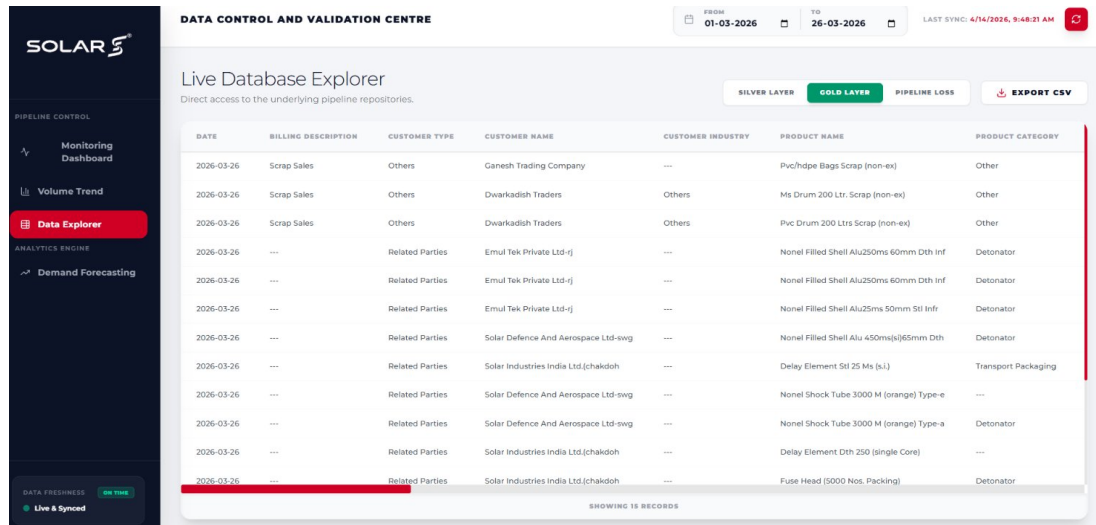


Figure 2.16: Live Database Explorer – Gold Layer View with Real Sales Records

Table 2.4: Software and Hardware Specifications

Component	Technology
Backend Framework	Python 3, Django, Django REST Framework (DRF)
Database	PostgreSQL (AWS RDS), Multi-Database Routing
Classic AI Models	Random Forest, LightGBM, XGBoost
Neural AI Models	TFT, N-BEATS
Frontend	React.js, TypeScript, Tailwind CSS
Visualization	Recharts, Three.js (WebGL)
Infrastructure	Multi-threaded ThreadPoolExecutor
Data Pipeline	Medallion Architecture (Bronze/Silver/Gold)
Evaluation Metrics	WAPE, R^2 Score, RMSE, MAPE
Processor	Intel i5 or above
RAM	8 GB minimum
Operating System	Windows / Linux
Cloud	Amazon Web Services (AWS RDS)

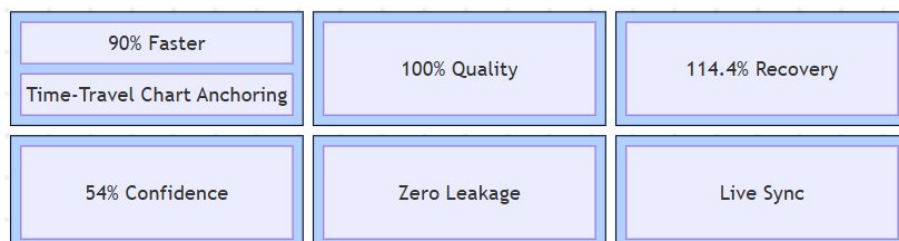


Figure 2.17: Key Performance Metrics of the Data Control and AI Demand Sensing Platform

Table 2.5: System Performance Evaluation Results

Test Case	Description	Result
TC1	Dashboard Loading Latency	10s → <1s (90% faster)
TC2	Identity Validation Score	100%
TC3	Data Uniqueness Score	100%
TC4	Best Model Confidence Score (LightGBM)	59%
TC5	Period Retention Rate	100%
TC6	Pipeline Loss (Duplicates removed)	0 Records
TC7	Forecast Horizons Supported	7, 15, and 30 Days
TC8	SLA Alert System	Live & Synced (Green)



Figure 2.18: System Volume Trend – Ingestion vs. Delivery Gap (March 2026)

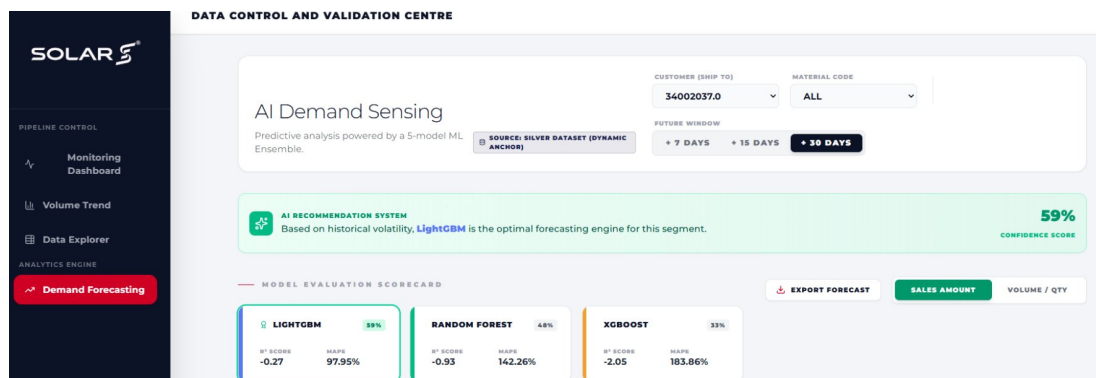


Figure 2.19: AI Demand Sensing – Model Evaluation Scorecard (LightGBM: 59% Confidence)

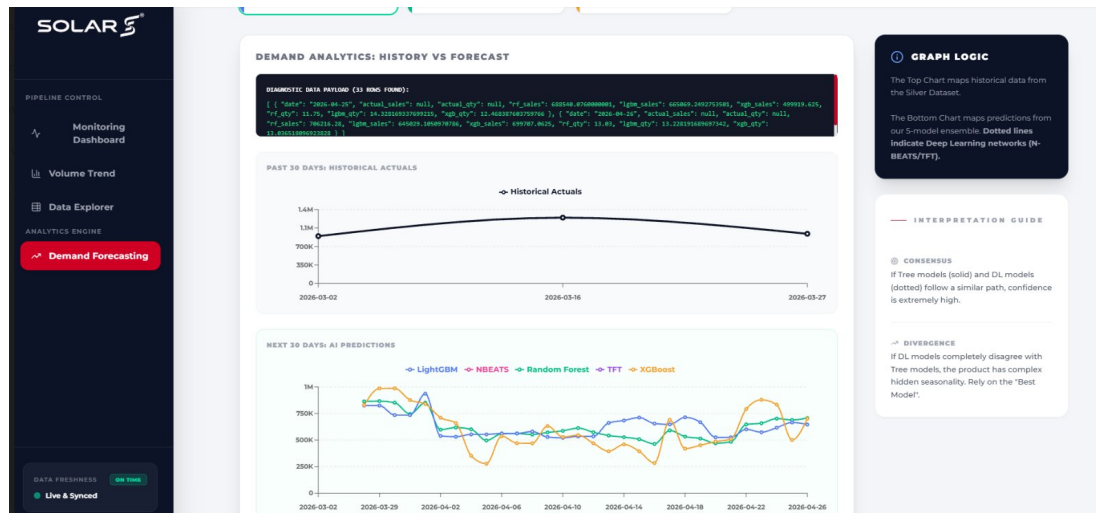


Figure 2.20: Demand Analytics: History vs. Forecast – All 5 Model Predictions

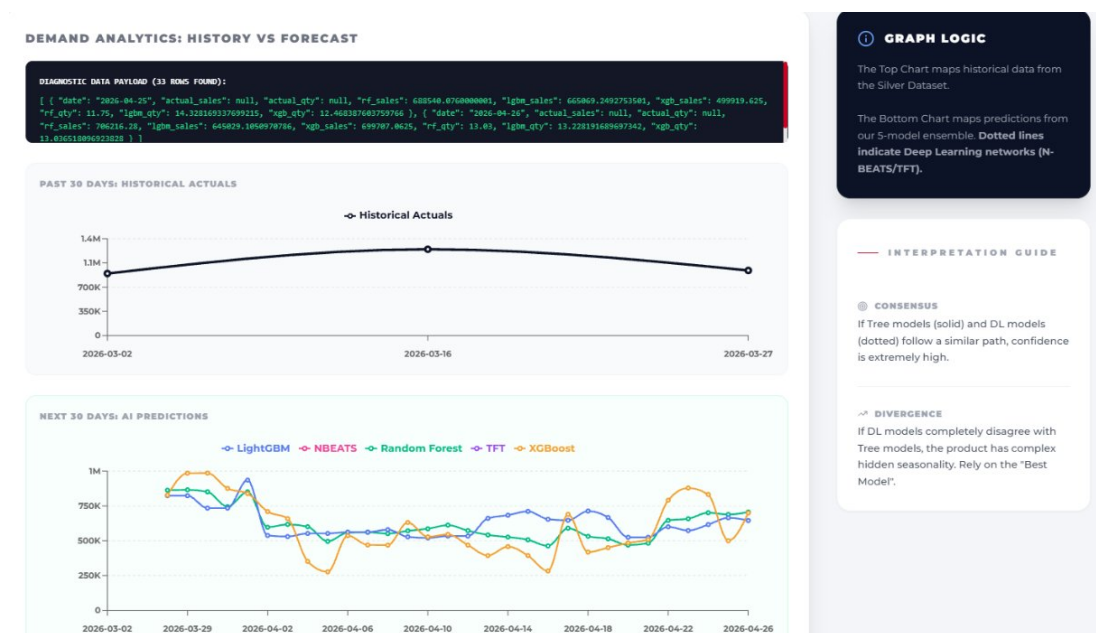


Figure 2.21: Demand Analytics: Zoomed View – Next 30 Days AI Predictions

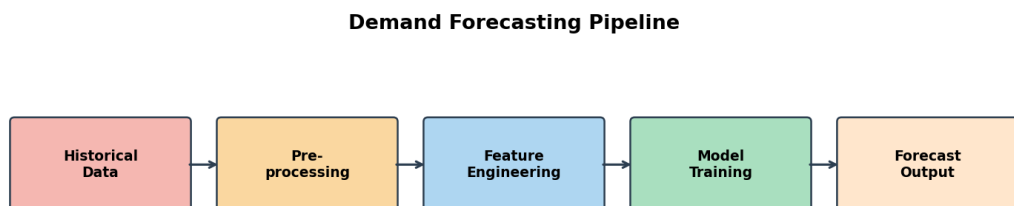


Figure 2.22: Demand Forecasting Pipeline



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FUTURE SCOPE AND ENHANCEMENTS

The current implementation of the Data Control and AI Demand Sensing Platform provides a strong operational foundation. Several enhancements are planned for future development phases, as summarized in Figure 2.23.

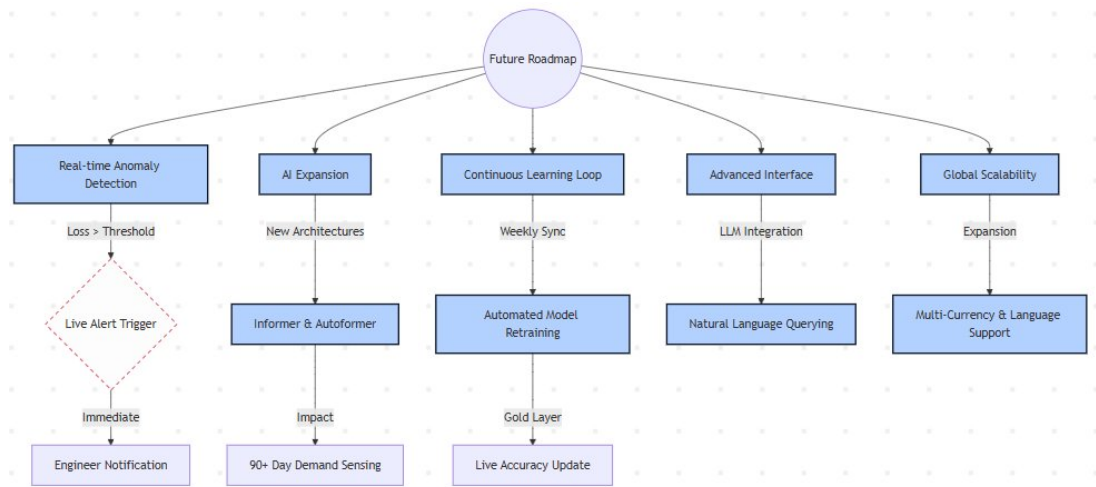


Figure 2.23: Future Development Roadmap

- 1. Real-Time Anomaly Detection Alerts:** Setting up instant push notifications to business managers if data loss exceeds a threshold during ERP sync, triggering live alert notifications to engineers for immediate resolution.
- 2. AI Expansion with Informer and Autoformer:** Adding next-generation neural architectures such as Informer and Autoformer to extend the forecast horizon to 90+ days, enabling longer-term strategic planning.
- 3. Continuous Learning Loop:** Creating an automated model retraining loop where all AI models retrain on newly available Gold Layer data every week, ensuring forecasting accuracy improves continuously with live accuracy updates.
- 4. Advanced Interface with LLM Integration:** Building a natural language query interface powered by an LLM so that non-technical managers can request sales forecasts and data summaries in plain English via natural language querying.
- 5. Global Scalability:** Expanding the platform to support multiple languages, currencies, and regional data formats to serve Solar Industries' international manufacturing plants in Africa, Southeast Asia, and the Middle East.

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APPENDIX A: INTERNSHIP COMPLETION CERTIFICATE

*[To be attached — Certificate from Solar Industries India Limited on company letterhead,
stating duration and project details, as per the format prescribed by SIT Nagpur.]*

APPENDIX B: PLAGIARISM REPORT



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Arushi Shivhare Internship_Report



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APPENDIX C: AI SIMILARITY REPORT



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Internship_Report



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The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

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Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

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